Practical – 2

Aim : String Validation using Finite Automata

1. String over 0 and 1 where every 0 immediately followed by 11.

01011

```
Enter number of input symbol: 2
Enter the 1 input Symbol : 0
Enter the 2 input Symbol : 1
Enter number of state: 4
Enter initial state: 1
Enter number of accepting state: 1
Enter the 1 accepting  State : 1
Transition table:
1 to 0 -> 2
1 to 1 -> 1
2 to 0 -> 4
2 to 1 -> 3
3 to 0 -> 4
3 to 1 -> 1
4 to 0 -> 4
4 to 1 -> 4
Input string: 01011
Invalid string.

--------------------------------
Process exited after 15.72 seconds with return value 0
Press any key to continue . . .
```

2. string over a b c, starts and end with same letter

   Cabc

```
Enter number of input symbol: 3
Enter the 1 input Symbol : a
Enter the 2 input Symbol : b
Enter the 3 input Symbol : c
Enter number of state: 7
Enter initial state: 1
Enter number of accepting state: 3
Enter the 1 accepting  State : 2
Enter the 2 accepting  State : 4
Enter the 3 accepting  State : 6
Transition table:
1 to a -> 2
1 to b -> 4
1 to c -> 6
2 to a -> 2
2 to b -> 3
2 to c -> 3
3 to a -> 2
3 to b -> 3
3 to c -> 3
4 to a -> 5
4 to b -> 4
4 to c -> 5
5 to a -> 5
5 to b -> 4
5 to c -> 5
6 to a -> 7
6 to b -> 7
6 to c -> 6
7 to a -> 7
7 to b -> 7
7 to c -> 6
Input string: cabc
Valid string.

--------------------------------
Process exited after 56.06 seconds with return value 0
Press any key to continue . . .
```

3. String over lower-case alphabet and digits, starts with alphabet only.

```
1009modi
Invalid String
```